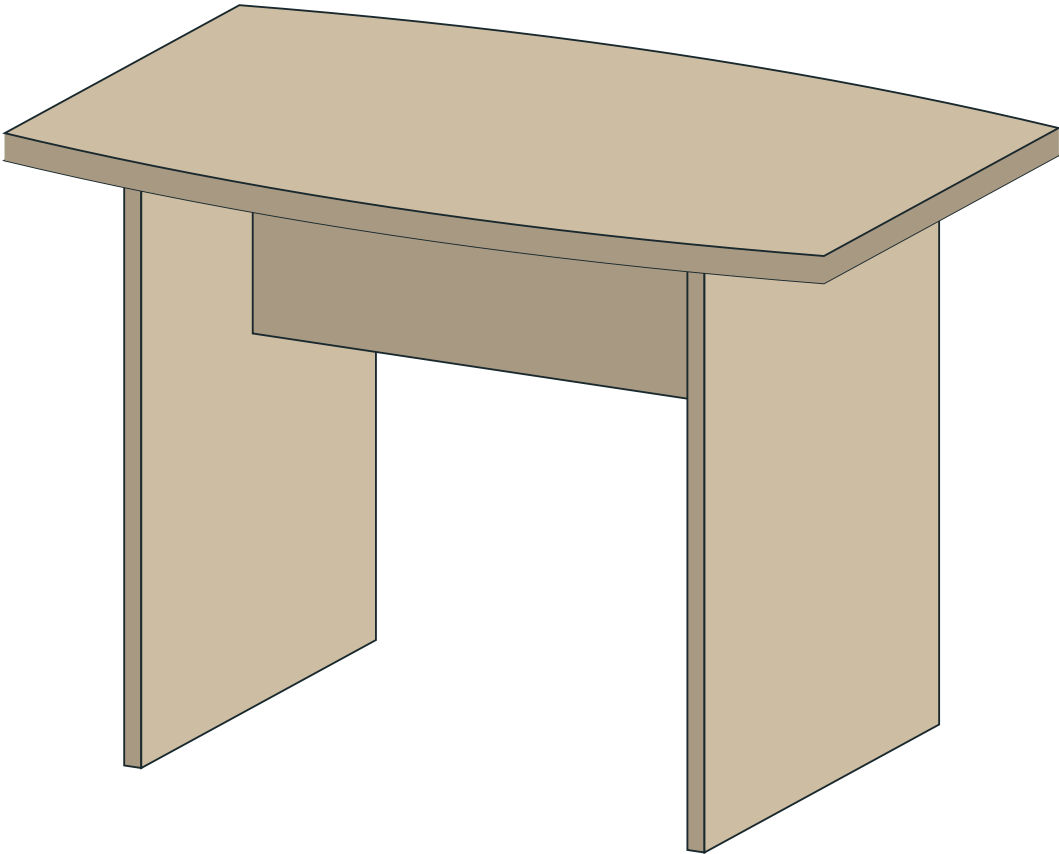


1 / Simple boat table

The easiest showcase: four boards, eight locks and six distinct assembly movements.

FINISHED ASSEMBLY



OVERALL ENVELOPE / X x Y x Z

1200 x 720 x 600 mm

Top width: 600 center; 500 at both ends.

Boat edges: $z = +/- [300 - 50(x/600)^2]$.

Top of bases and stretcher: Y=688.

Top: 32 thick; both bases: 25 thick.

THE CONNECTION IS THE FEATURE

Each green U-shaped fitting has two barbed posts fixed inside its carrier. Its exposed bridge enters a route in the other board, then slides beneath the retaining shoulders.

Sheets 2-5 locate every fitting and define the two distinct movements. Sheet 6 is the supplied Mini routing reference.

4 boards / 8 channel locks / 6 movements

ID	COMPONENT	WORLD SIZE X x Y x Z	QTY	FITTING ROLE
P01	Boat tabletop	1200 x 32 x 600	1	Receives 4 locks underneath
P02	Left base	25 x 688 x 500	1	Carries 2 top locks; receives 2
P03	Right base	25 x 688 x 500	1	Carries 2 top locks; receives 2
P04	Center stretcher	800 x 250 x 25	1	Carries 2 locks in each end

Exploded view / all eight fittings

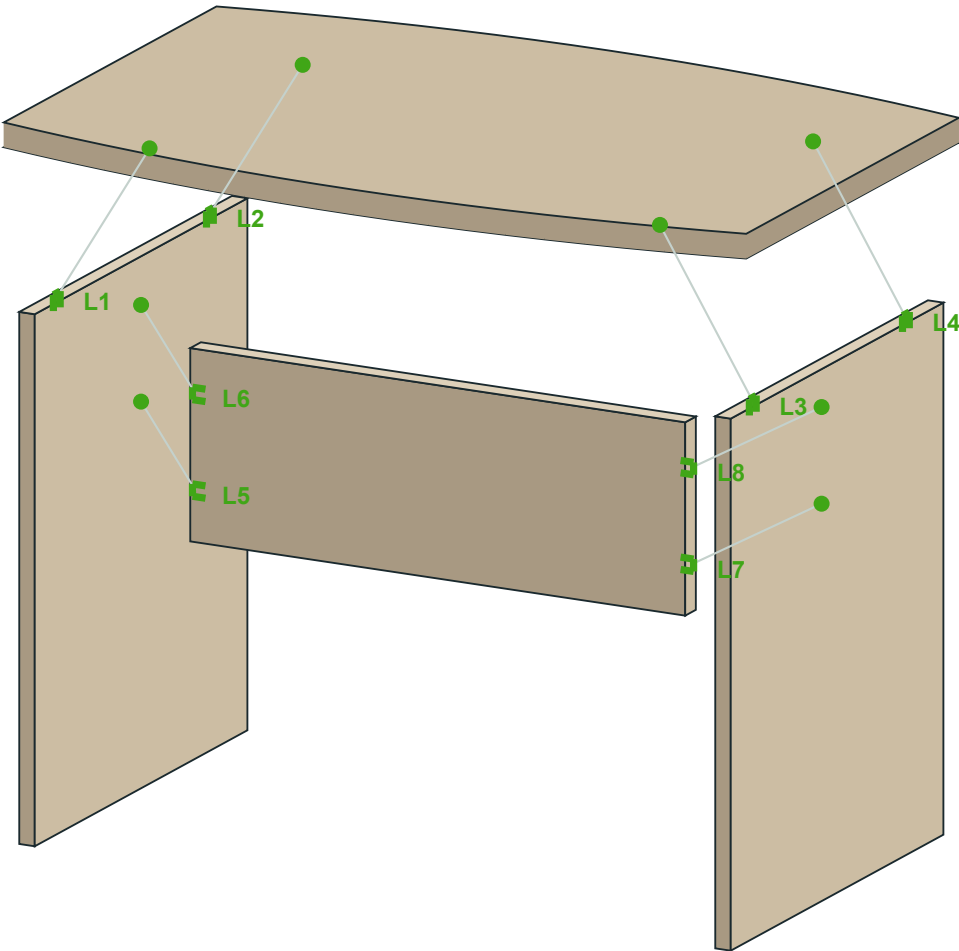
Separated for readability only. Use sheet 5 for the actual assembly movements.

P01 / BOAT TABLETOP

Receives 4 locks underneath.

P04 / CENTER STRETCHER

Carries 2 locks in each end.



P02 / LEFT BASE

Carries 2 top locks; receives 2.

P03 / RIGHT BASE

Carries 2 top locks; receives 2.

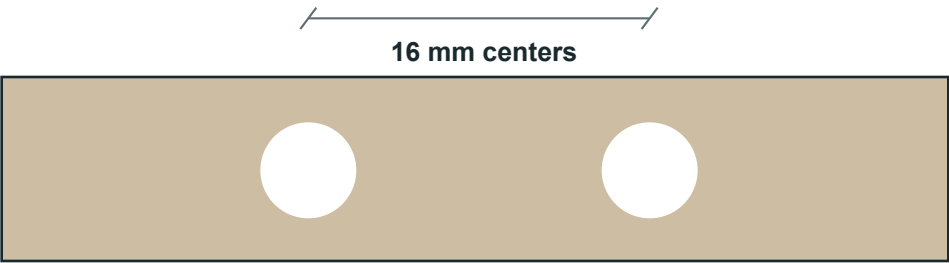
ONE LOCK = ONE U-SHAPED BRIDGE + TWO BARBED POSTS

8 physical locks require 16 carrier bores and 8 receiver routes. The posts stay embedded in the carrier; the other board receives the exposed bridge. Green dots mark the matching receiver anchors. L1-L8 match the location schedule on sheet 4.

Inside the Lockdowel connection

Detail diagrams are schematic. Dimensions marked "concept" define the film, not production tooling.

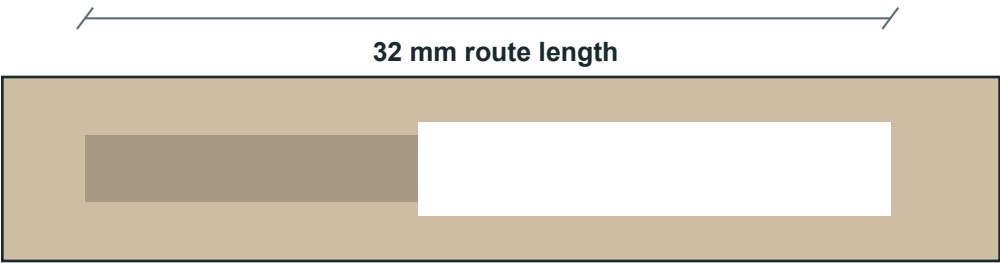
A / TWO BORES IN THE CARRIER



2 x diameter 5 mm, 16 mm deep

Posts point inward; the bridge remains outside.

B / ROUTE IN THE RECEIVER

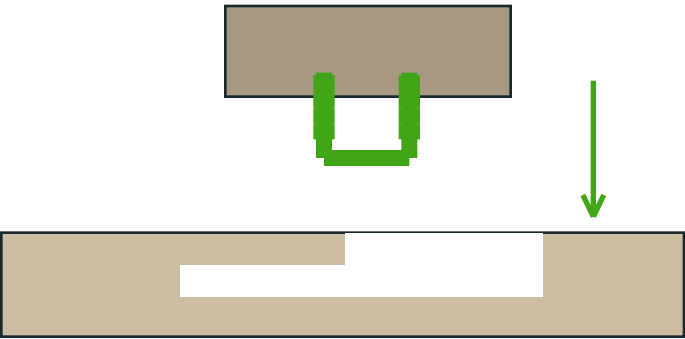


Narrow capture region

Wide entry region

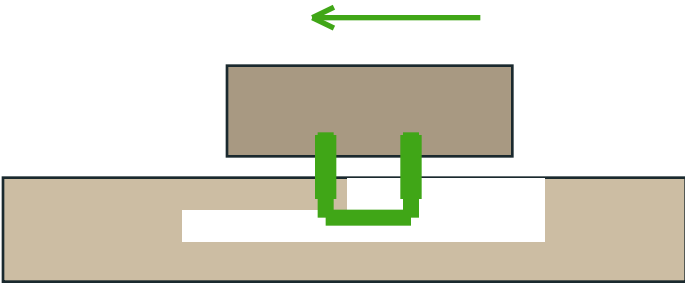
Shallow depth 3; deep depth 6 to 6.3 mm. See sheet 6.

C / ENTER FIRST. SLIDE SECOND.



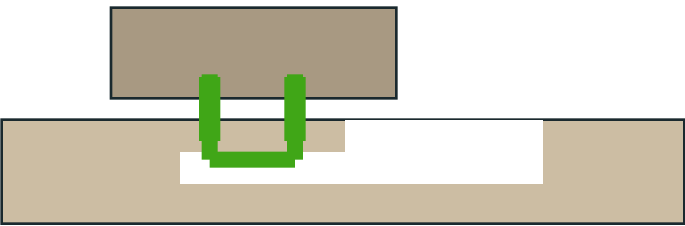
1 / APPROACH

Move perpendicular to the receiver. The fitting travels with its carrier.



2 / BRIDGE ENTERED

The bridge is in the wide opening. Now move along the slot, not into it again.



3 / SLID UNDER THE LIP

The retaining shoulder captures the bridge. The two posts remain in the carrier.

CONCEPT PARAMETERS: 21 mm bridge length; 6.35 mm undercut width; 35 mm approach clearance; 14 mm locking travel. Exact SKU cross-section and machining tolerances need engineering confirmation. Use the original routing sheet for source dimensions.

Component positions / exact fitting anchors

Coordinates: X left/right, Y up, Z front/back. Front is negative Z. Origin: center of the floor footprint.

ID	COMPONENT	WORLD SIZE X x Y x Z	FINAL CENTER (X, Y, Z)
P01	Boat tabletop	1200 x 32 x 600	0, 704, 0
P02	Left base	25 x 688 x 500	-412.5, 344, 0
P03	Right base	25 x 688 x 500	412.5, 344, 0
P04	Center stretcher	800 x 250 x 25	0, 563, 0

EACH ROW IS ONE U-SHAPED LOCK

LOCK	CARRIER	RECEIVER	ANCHOR P (X, Y, Z)	OUTWARD C	SLOT AXIS U
L1	P02	P01	-412.5, 688, -180	0, 1, 0	0, 0, 1
L2	P02	P01	-412.5, 688, 180	0, 1, 0	0, 0, 1
L3	P03	P01	412.5, 688, -180	0, 1, 0	0, 0, 1
L4	P03	P01	412.5, 688, 180	0, 1, 0	0, 0, 1
L5	P04	P02	-400, 500, 0	-1, 0, 0	0, -1, 0
L6	P04	P02	-400, 625, 0	-1, 0, 0	0, -1, 0
L7	P04	P03	400, 500, 0	1, 0, 0	0, 1, 0
L8	P04	P03	400, 625, 0	1, 0, 0	0, 1, 0

BORE AND ROUTE LOCATION RULE

P is the final seated anchor on the two mating faces. Carrier bore entries are P - 8U and P + 8U; bore 16 mm inward along -C. For the concept, the receiver route extends from P - 9U to P + 23U; its wide entry is toward +U. Cut into the receiver along +C. U is the slot orientation, not necessarily the moving board direction: that depends on which board moves (sheet 5).

Six movements / a complete connection film

Start with P02 supported upright. Preinstall all eight U locks in their specified carrier boards.

STEP	MOVING COMPONENT	ACTION	FROM CENTER	TO CENTER
1	P04 / Center stretcher	Normal entry / 35 mm	35, 549, 0	0, 549, 0
2	P04 / Center stretcher	Lock slide / 14 mm	0, 549, 0	0, 563, 0
3	P03 / Right base	Normal entry / 35 mm	447.5, 330, 0	412.5, 330, 0
4	P03 / Right base	Lock slide / 14 mm	412.5, 330, 0	412.5, 344, 0
5	P01 / Boat tabletop	Normal entry / 35 mm	0, 739, -14	0, 704, -14
6	P01 / Boat tabletop	Lock slide / 14 mm	0, 704, -14	0, 704, 0

DIRECTION SCHEDULE

STEP	TRANSLATION (dX,dY,dZ)	DIRECTION	MECHANISM
1	-35, 0, 0	left / 35 mm	Exposed bridge enters the routed opening
2	0, 14, 0	up / 14 mm	Bridge engages beneath retaining shoulders
3	-35, 0, 0	left / 35 mm	Exposed bridge enters the routed opening
4	0, 14, 0	up / 14 mm	Bridge engages beneath retaining shoulders
5	0, -35, 0	down / 35 mm	Exposed bridge enters the routed opening
6	0, 0, 14	back / 14 mm	Bridge engages beneath retaining shoulders

FILM REQUEST: Show all four boards and all eight fittings. Keep each U lock attached to its carrier. Use a fixed camera during each movement and a clear connector cutaway during entry and sliding. Finish on the complete assembly. Do not add fittings or substitute a generic exploded overview for the locking motion.

Original Mini routing reference

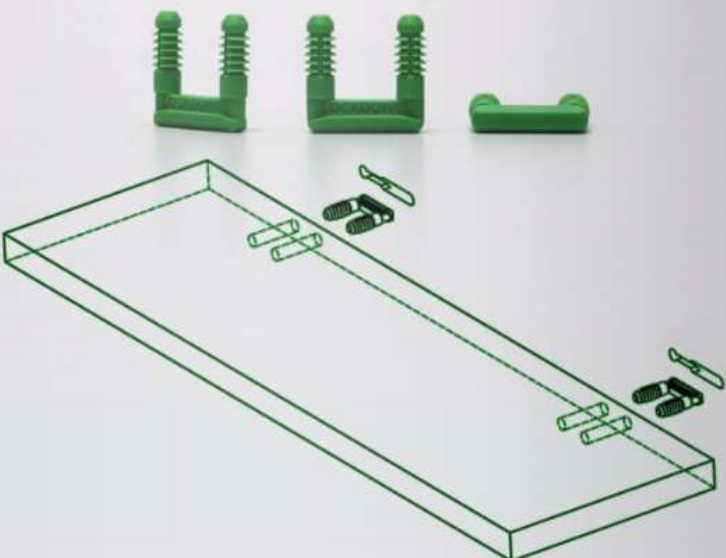
Source: supplied "LD 5 inch E900BP-mini drawer box details (1).pdf", page 11. Reference only; no drawer parts are included.

E900BPMini Mini Nylon Barbed Channel Lock

Specification

- Removable Sturdy Joint
- Pull Force: 70+ lbs
- Sheer Force: Over 50 lbs
- Requires CNC routing capability and edge boring machine
- Best used in Particle Board, Plywood, MDF, and selected plastic materials (Min. Thickness: 1/2")

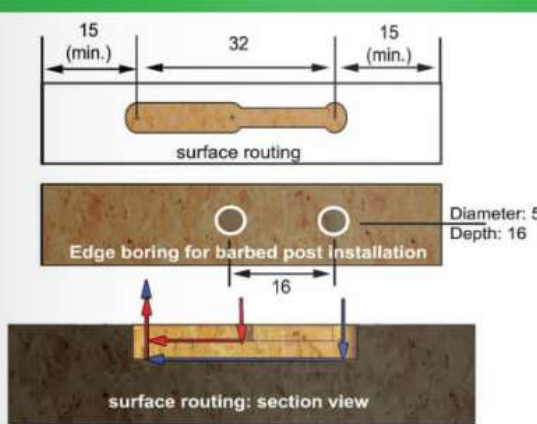
- Drilling: Ø 5 mm and 16 mm spacing
- Material: Nylon 6-6



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Patent Pending

Mini Barbed Channel Lock Routing Instructions



15 (min.) 32 15 (min.)

surface routing

Edge boring for barbed post installation

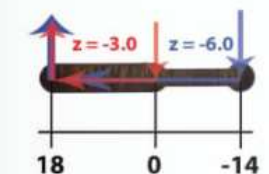
Diameter: 5*
Depth: 16

16

surface routing: section view

Plunge Cut: Z: -3.0 Slot Cut: X: 0 to 18
Plunge Cut: Z: -6.0 Key Slot: X: -14 to 18
(Allowable Depth Range is -6.0 to -6.3)

Route RED first. Then BLUE.



z = -3.0 z = -6.0

18 0 -14

• All measurements are in mm.
• If using MDF, replace 5mm Drill Bit with 5.2mm Drill Bit.

11

Patent Pending